



Pirfenidone improves the survival of patients with idiopathic pulmonary fibrosis hospitalized for acute exacerbation

Andrea Vianello, Beatrice Molena, Cristian Turato, Fausto Braccioni, Giovanna Arcaro, Luciana Paladini, Margherita Andretta & Marina Saetta

To cite this article: Andrea Vianello, Beatrice Molena, Cristian Turato, Fausto Braccioni, Giovanna Arcaro, Luciana Paladini, Margherita Andretta & Marina Saetta (2019) Pirfenidone improves the survival of patients with idiopathic pulmonary fibrosis hospitalized for acute exacerbation, Current Medical Research and Opinion, 35:7, 1187-1190, DOI: [10.1080/03007995.2019.1565530](https://doi.org/10.1080/03007995.2019.1565530)

To link to this article: <https://doi.org/10.1080/03007995.2019.1565530>



Published online: 24 Jan 2019.



Submit your article to this journal [↗](#)



Article views: 710



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 3 View citing articles [↗](#)

BRIEF REPORT



Pirfenidone improves the survival of patients with idiopathic pulmonary fibrosis hospitalized for acute exacerbation

Andrea Vianello^a, Beatrice Molena^a, Cristian Turato^b, Fausto Braccioni^a, Giovanna Arcaro^a, Luciana Paladini^a, Margherita Andretta^c and Marina Saetta^a

^aDepartment of Cardiological, Thoracic and Vascular Sciences, University of Padova, Padova, Italy; ^bVeneto Institute of Oncology IOV – IRCCS, Padova, Italy; ^cHealth Technology Assessment Unit, Azienda Zero, Padova, Italy

ABSTRACT

Objective: To examine the effect of pirfenidone on the survival of patients hospitalized due to acute exacerbation of idiopathic pulmonary fibrosis (AE–IPF).

Methods: The outcomes of 11 consecutive AE–IPF patients who were receiving pirfenidone treatment when they were admitted to a respiratory intensive care unit (RICU) for acute respiratory failure (ARF) (treatment group) were retrospectively compared with those of 9 patients who were not on pirfenidone treatment at admission (control group). The study's primary outcome measure was survival following RICU admission; the patients' mortality rate and the length of time spent in the RICU were also assessed.

Results: The treatment group had significantly longer survival than the control group (median survival time: 137.0 [95% CI, 39.0–373.0] versus 16.0 [95% CI, 14.0–22.0] days; $p = .0009$); the hazard ratio for death was 0.2896 (95% CI, 0.09541–0.8791). The treatment group also tended to have a lower RICU mortality rate (3/11 vs. 7/9; $p = .0698$).

Conclusions: Pirfenidone significantly improved survival in IPF patients hospitalized for severe acute exacerbation compared to controls.

ARTICLE HISTORY

Received 28 October 2018
Revised 21 December 2018
Accepted 3 January 2019

KEYWORDS

Idiopathic pulmonary fibrosis; pirfenidone; acute exacerbation; acute respiratory failure

Introduction

A chronic interstitial lung disease of unknown etiology, idiopathic pulmonary fibrosis (IPF) generally leads to a progressive decline in respiratory function and early mortality. Some IPF patients can experience acute exacerbation (AE) leading to severe acute respiratory failure (ARF) and the need for hospitalization¹. A recently published international working group report defined AE–IPF as “an acute, clinically significant respiratory deterioration characterized by evidence of new widespread alveolar abnormality”².

Approved for use in patients with IPF, pirfenidone has been found to delay disease progression, as defined by patients' 1 year decline in forced vital capacity (FVC)^{3–5}. It has also been reported to reduce decline in exercise capacity, as defined by the 6-minute-walk distance (6MWD), and to improve patients' one-year progression-free survival⁴.

Although an analysis of pooled patient data from three phase III clinical trials uncovered that pirfenidone was associated with a significant reduction in the risk of respiratory-related hospitalization and of death after hospitalization, its effect on the outcomes of patients hospitalized for AE–IPF is still uncertain⁶. The current study set out to retrospectively analyze the clinical outcomes of patients admitted to an intensive care unit due to severe AE–IPF. The data on patients who were taking pirfenidone at the time they were

admitted (the treatment group) was compared with that of patients who were not on pirfenidone therapy (the control group).

Methods

This retrospective study, which was conducted in a tertiary teaching hospital located in Northeast Italy, was approved by the facility's Institutional Review Committee. The participants signed informed consent forms releasing their medical records for review.

Patients

The medical records of IPF patients with AE who were admitted to the four-bed respiratory intensive care unit (RICU) of the University of Padua Medical Center between 1 January 2010 and 31 December 2017 for ARF were collected and reviewed. The diagnosis of IPF was made using the multidisciplinary approach approved by the criteria proposed by the ATS/ERS consensus statement¹. The diagnosis of AE was made in accordance with the recent recommendations of an expert panel² and did not include lung biopsy.

The baseline demographic, clinical, pulmonary and cardiac function data of the patients meeting the study's eligibility

Table 1. Patients' baseline demographic, clinical, pulmonary and cardiac function characteristics.

	Treatment group (n = 11)	Control group (n = 9)	p value
Age (years), median (range)	61.9 (51.6–78.4)	73.8 (50.1–87.3)	.1194
Gender (males/females)	9/2	6/3	.6168
BMI (kg/m ²), median (range)	26.42 (18.73–31.24)	22.91 (19.6–29.4)	.3051
Number of previous smokers	9 (81.8%)	7 (77.8%)	>.9999
Number of pts with UIP pattern on HRCT	5	4	>.9999
Number of pts who underwent lung biopsy	4	4	>.9999
Length of time from diagnosis to RICU admission (months), median (range)	27.02 (2.10–60.13)	35.97 (22.72–129.07)	.1351
Number of hospitalizations earlier in the year, median (range)	0 (0–2)	0 (0–2)	>.9999
Number of co-morbidities, median (range)	2 (0–5)	3 (0–4)	.6445
Number of pts with previous cardiac disease	2 (18.2%)	5 (55.6%)	.1596
Number of pts previously administered corticosteroids	1	4	.1273
Number of pts previously administered other immunosuppressive therapy	0	3	.0737
Number of pts previously administered long-term oxygen therapy	8 (72.7%)	6 (66.7%)	>.9999
Number of pts previously administered long-term NIV	1 (9.1%)	1 (11.1%)	>.9999
Number of pts on waiting list for lung transplantation	3 (27.3%)	3 (33.3%)	>.9999
FVC (L), median (range)	1.82 (1.11–2.75)	1.64 (1.20–2.33)	.6877
FVC (%), median (range)	55.00 (36.00–101.00)	41.50 (21.00–78.00)	.3657
FEV ₁ (L), median (range)	1.70 (1.09–2.66)	1.41 (1.12–2.03)	.4510
FEV ₁ (%), median (range)	68.00 (41.00–120.00)	47.50 (25.00–73.00)	.1911
DLCO (ml/min/mmHg), median (range)	5.63 (2.33–7.33)	5.05 (2.91–7.90)	.8973
DLCO (%), median (range)	28.00 (11.0–49.0)	16.50 (6.0–45.0)	.1407
Length of time from PFT to RICU admission (months), median (range)	1.85 (0.23–13.43)	17 (1.1–22.87)	.0864
Number of pts with PH	9 (81.8%)	5 (55.6%)	.6026
GAP index, median (range)	5.0 (3.0–6.0)	4.5 (2.0–7.0)	.6792

Abbreviations. BMI, Body mass index; DLCO, Diffusing capacity for carbon monoxide; FVC, Forced vital capacity; FEV₁, Forced expiratory volume in the 1st second; LVEF, Left ventricular ejection fraction; NIV, Non-invasive ventilation; HRCT, High resolution chest tomography; PH, Pulmonary hypertension; Pts, Patients; PFT, Pulmonary function test; RICU, Respiratory intensive care unit; UIP, Usual interstitial pneumonia.

Table 2. Patients' clinical and laboratory data at RICU admission.

	Treatment group (n = 11)	Control group (n = 9)	p value
Respiratory rate (breaths/min), median (range)	25 (16–42)	32 (16–42)	.3414
Heart rate (beats/min), median (range)	104 (58–130)	100 (85–170)	.5943
GCS, median (range)	15 (12–15)	15 (14–15)	.9513
Number of pts with fever (temperature >38 °C)	1 (9.1%)	0 (0.0%)	>.9999
Number of pts with leukocytosis (WBC >12,000 × 10 ⁶ /L)	5 (45.5%)	6 (66.7%)	.4058
PaO ₂ ^a (mmHg), median (range)	79.0 (39.0–258.7)	68.3 (34.0–89.3)	.2706
PaCO ₂ (mmHg), median (range)	40.8 (27.5–82.3)	38.6 (25.3–92.0)	.7040
Arterial pH, median (range)	7.44 (7.26–7.41)	7.44 (7.39–7.51)	.7039
SaO ₂ (%), median (range)	92.0 (70.0–98.0)	93.0 (60.5–95.0)	.4704
PaO ₂ /FiO ₂ , median (range)	145.0 (46.0–289.0)	130.5 (73.0–268.0)	.6203
Number of pts with PaO ₂ /FiO ₂ level ≤100	2	2	>.9999
NT-proBNP (pg/mL), median (range)	259.0 (44.0–6309.0)	2641.0 (663.0–6211.0)	.1745
Number of pts with abnormal NT-proBNP level	4	7	.0587
Cardiac troponin (ng/mL), median (range)	2.0 (0.017–203.0)	0.168 (0.08–1799.0)	.4875
CRP (μg/mL), median (range)	28.4 (8.1–140.0)	87.0 (5.3–300.0)	.4033
Number of pts with CRP level >100 μg/mL	1 (9.1%)	3 (33%)	.2848
Number of pts with D-dimer level ≥0.4 mcg/mL	6	8	.1571
APACHE II score, median (range)	20 (5–33)	17 (11–26)	.4472

^aWith supplemental oxygen.

Abbreviations. APACHE, Acute Physiology and Chronic Health Evaluation; CRP, C-reactive protein; GCS, Glasgow Coma Scale; NT-proBNP, Prohormone of B-type natriuretic peptide; PaO₂/FiO₂, Arterial oxygen tension to inspired oxygen fraction ratio; PCT, Procalcitonin; SaO₂, Arterial oxygen saturation; WBC, White blood cells.

criteria was reviewed (Table 1). All clinical parameters and biological markers at the time they were admitted to the RICU were also analyzed (Table 2). Data on patients who underwent lung transplant (LT) later was not analyzed.

Treatment

Pirfenidone was regularly taken at the standard dose of 801 mg (three capsules) three times daily by all the patients in the treatment group. All side-effects potentially linked to the drug were recorded.

All the patients were prescribed all appropriate supportive measures for ARF including conventional and/or high-flow nasal cannula (HFNC) oxygen therapy, and/or non-invasive ventilation (NIV). Utilization of elective intubation

and invasive mechanical ventilation following NIV failure was not contemplated by standard protocol in our RICU except for those patients on a waiting-list for LT. High-dose corticosteroid therapy (1 mg/kg/day) and broad-spectrum antibiotic regimens were administered to all the patients throughout their stay in the RICU.

Study endpoints and statistical analysis

The patients were divided into two groups depending on their therapeutic regimen. The first (treatment group) was made up of those individuals who were taking pirfenidone at the time they were admitted to the RICU; the second (control group) was made of those subjects who were not on pirfenidone treatment.

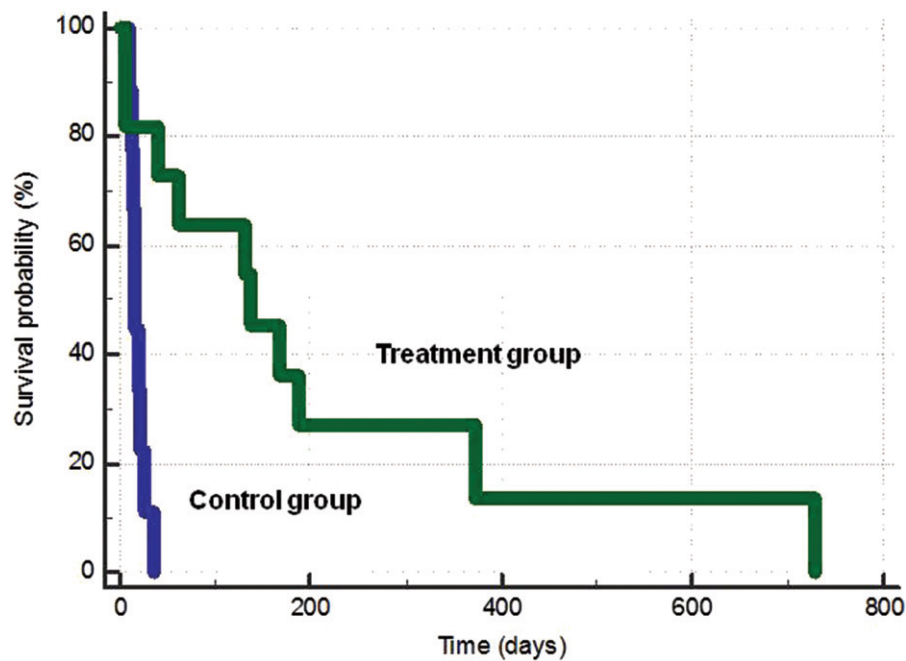


Figure 1. Kaplan–Meier estimates of survival function after RICU admission, stratified according to the group of origin.

The patients' outcomes, as deduced from their clinical status parameters at the time they were discharged from the RICU and at the end of the follow-up period, and the length of time the patients remained in the RICU were analyzed. The duration of the follow-up period was defined as the time from RICU discharge to the last observation available for each patient. The study's primary endpoint was survival following RICU admission. The mortality rate during the stay in the RICU and the length of time the patient remained in the RICU were the secondary endpoints.

The results are expressed, as appropriate, as mean values $\pm$ SD, medians and percentages. The Kolmogorov–Smirnov test was used to check the normality of the data distribution. Continuous variables were compared, depending on the normality of the distributions, using Student's *t* test or the Mann–Whitney *U* test. Categorical variables were compared, as appropriate, using the chi-squared test or Fisher's exact test. Survival from the time of admission to the RICU was calculated using the Kaplan–Meier method; the log-rank test was used to compare the survival curves of the treatment and control groups.

All statistical calculations were carried out using Medall Statistical Software (Ostend, Belgium). A bilateral *p* value $< .05$ was considered statistically significant for all comparisons.

Results

The twenty-two patients who were admitted to our RICU with a diagnosis of AE-IPF during the study period (1 January 2010–31 December 2017) were considered eligible to participate in our retrospective study. Data on the two patients who underwent LT at a later date was not, however, included in the analysis. Eleven of the remaining 20 patients were regularly taking standard doses of pirfenidone therapy

(treatment group); 9 were not receiving the medication (control group). The median duration of pirfenidone treatment was 22.55 (0.79–50.10) months. No adverse drug effect potentially linked to pirfenidone was reported. The reasons the control patients were not prescribed pirfenidone prior to admission include the following: the drug was not yet on the market (3 cases; pirfenidone was, in fact, approved for IPF treatment by the regional health authority on 1 August 2013); the indications for the drug did not include the age group into which the patient fell (3 cases); poor compliance (2 cases); declined treatment (1 case). There were no significant differences in the baseline demographic, clinical, pulmonary/cardiac function, or radiologic characteristics (outlined in Table 1) between the two patient groups. The clinical and laboratory values of patients in the treatment group were similar to those of the control group at the time of RICU admission, although more patients in the latter group had above normal plasma N-terminal pro-brain natriuretic peptide (NT-proBNP) levels (Table 2). None of the patients in either the treatment or the control group received recombinant human soluble thrombomodulin (rhTM) during hospitalization.

The survival rate of the patients in the treatment group was 70.0% ($\pm 10.2\%$) at 15 days, 45.0% ($\pm 11.1\%$) at 30 days, 35.0% ($\pm 10.7\%$) at 60 days and 7.5% ($\pm 6.6\%$) at 370 days (maximum length of the follow-up period). The stratified log-rank test showed that the patients in the treatment group survived for a significantly longer time than those in the control group (median survival time: 137.0 [95% CI, 39.0–373.0] versus 16.0 [95% CI, 14.0–22.0] days; $p = .0009$) (Figure 1); their hazard ratio for death was 0.2896 (95% CI, 0.09541–0.8791).

The patients in the treatment group tended to have a lower RICU mortality rate than the control group (3/11 vs.

7/9; $p = .0698$). Finally, the length of time in the RICU was similar in the two groups (12.0 [95% CI, 5.0–39.0] vs. 17.0 [95% CI, 10.0–25.0] days; $p = .1487$).

Discussion

To our knowledge, this is the first study to evaluate the effect of pirfenidone on the clinical outcomes of patients affected with AE-IPF being admitted to a RICU. When the data of the previously treated and untreated AE-IPF patients with similar clinical features and respiratory status parameters were compared, it was found that pirfenidone had a beneficial effect on the former's survival and mortality during their stay in a RICU.

Translating to AE-IPF the Berlin criteria utilized to categorize subjects with ARDS according to $\text{PaO}_2/\text{FiO}_2$ ratio⁷, our patients can be classified as having either moderate (16 cases) or severe (4 cases) acute respiratory distress at RICU entry; the $\text{PaO}_2/\text{FiO}_2$ ratio was calculated during application of low PEEP level (5 cm H_2O) in those cases who received NIV. In line with previous data reporting that a high D-dimer level is associated with an increased risk of developing AE⁸, 14 out of 20 patients showed an abnormally elevated D-dimer level (≥ 0.4 mcg/mL) at admission.

The mechanism of action by which pirfenidone therapy was associated with an approximate one-third reduction in the risk of death for the treated, hospitalized AE-IPF patients is as yet unexplained. Analysis of study data did, however, uncover one intriguing finding: the treatment group had (slightly) significantly fewer patients with abnormal plasma NT-proBNP levels at the time of RICU admission (4/11 vs. 7/9; $p = .0587$). Hypothetically, we could argue that pirfenidone has the ability to protect against the loss of left ventricular (LV) function, thereby limiting the risk of acute heart failure and death in the event of acute respiratory distress. In accordance with this hypothesis, apart from right ventricular (RV) dysfunction, IPF patients may also exhibit LV diastolic impairment caused by cardiac remodeling which may affect disease progression and prognosis⁹. Pirfenidone may hypothetically prevent pathological remodeling by modulating morphological and functional changes¹⁰.

The study presents inevitable limitations: first and foremost, the limited number of patients enrolled. Undeniably, all clinical studies focusing on patients with rare diseases such as IPF are afflicted with this limitation¹¹. Its second limitation is its retrospective nature which may have caused selection bias.

Conclusion

Despite these limitations, the study demonstrates that pirfenidone therapy significantly improves survival in IPF patients hospitalized for severe acute exacerbation. Larger studies are, of course, warranted to verify these findings.

Transparency

Declaration of funding

The study has not received any specific grants from funding agencies in the public, commercial or not-for-profit sectors. No study sponsor was involved in designing or conceptualizing the study, in collecting, analyzing/interpreting the data, in drafting the manuscript or in the decision to submit the manuscript for publication.

Author contributions: A.V.: study design, manuscript preparation; B.M.: data collection and analysis; C.T.: data collection and analysis; F.B.: study conduct; G.A.: study conduct; L.P.: study conduct; M.A.: study design; M.S.: study design, revision of the paper. All the authors have read and agreed on the contents of the paper and agree to be accountable for all aspects of the work.

Declaration of financial/other relationships

No potential conflict of interest was reported by the authors. *CMRO* peer reviewers on this manuscript have no relevant financial or other relationships to disclose.

Acknowledgements

No assistance in the preparation of this article is to be declared.

References

- [1] Raghu G, Collard HR, Egan JJ, et al. An official ATS/ERS/JRS/ALAT statement: idiopathic pulmonary fibrosis: evidence-based guidelines for diagnosis and management. *Am J Respir Crit Care Med*. 2011;183(6):788–824.
- [2] Collard HR, Ryerson CJ, Corte TJ, et al. Acute exacerbation of idiopathic pulmonary fibrosis. An International Working Group Report. *Am J Respir Crit Care Med*. 2016;194(3):265–275.
- [3] Noble PW, Albera C, Bradford WZ, et al. Pirfenidone in patients with idiopathic pulmonary fibrosis (CAPACITY): two randomised trials. *Lancet*. 2011;377(9779):1760–1769.
- [4] King TE Jr, Bradford WZ, Castro-Bernardini S, et al. A phase 3 trial of pirfenidone in patients with idiopathic pulmonary fibrosis. *N Engl J Med*. 2014;370(22):2083–2092.
- [5] Karimi-Shah BA, Chowdhury BA. Forced vital capacity in idiopathic pulmonary fibrosis – FDA review of pirfenidone and nintedanib. *N Engl J Med*. 2015;372(13):1189–1191.
- [6] Ley B, Swigris J, Day BM, et al. Pirfenidone reduces respiratory-related hospitalizations in idiopathic pulmonary fibrosis. *Am J Respir Crit Care Med*. 2017;196(6):756–761.
- [7] Ranieri VM, Rubenfeld GD, Thompson BT, et al. Acute respiratory distress syndrome: the Berlin Definition. *JAMA*. 2012;307:2526–2533.
- [8] Ishikawa G, Acquah SO, Salvatore M, et al. Elevated serum D-dimer level is associated with an increased risk of acute exacerbation in interstitial lung disease. *Respir Med*. 2017;128:78–84.
- [9] Papadopoulos CE, Pitsiou G, Karamitsos TD, et al. Left ventricular diastolic dysfunction in idiopathic pulmonary fibrosis: a tissue Doppler echocardiographic [corrected] study. *Eur Respir J*. 2008; 31(4):701–706.
- [10] Avila G, Osornio-Garduño DS, Ríos-Pérez EB, et al. Functional and structural impact of pirfenidone on the alterations of cardiac disease and diabetes mellitus. *Cell Calcium*. 2014;56(5):428–435.
- [11] Kuerner T. Essential rules and requirements for global clinical trials in rare lung diseases: a sponsor's standpoint. *Respir Investig*. 2015;53(1):2–6.